

nginx -> Envoy Gateway Migration Lab

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Problem: migrating 100 nginx Ingresses to Envoy Gateway — each with different annotations,
with no human watching dashboards, and with sub-10-second rollback on breach.

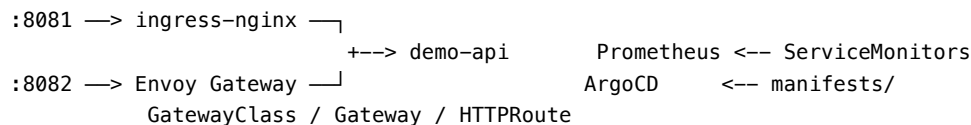
What Was Built

A production-grade Kubernetes lab that solves two concrete problems at the 100-endpoint scale:

- **Automated rollback** — a CronJob queries Prometheus every 60 s; on breach it rolls back every canary HTTPRoute fleet-wide with no human in the loop
- **Annotation translation** — ingress2gateway (official Kubernetes SIG tool) + per-Ingress policy stub generation converts the full annotation set to Gateway API CRDs automatically

Stack: k3d · Envoy Gateway v1.2 · ingress-nginx · kube-prometheus-stack · ArgoCD · Gateway API

Architecture



Both stacks serve the same backend simultaneously. Canary = backendRefs weight split inside one HTTPRoute. No DNS changes during ramp.

Automated Rollback

Core insight: at 100 endpoints you cannot watch per-route dashboards. Rollback must be metric-driven and fleet-scoped.

canary-watchdog CronJob — fires every 60 s, two breach conditions:

```
envoy 5xx / total > 5%          -> rollback
p99 downstream latency > 2000ms -> rollback
```

All HTTPRoutes labeled migration=canary are targeted in a single loop — 100 routes, ~5 to 10 seconds:

```
kubectl get httproute -A -l migration=canary ... | while read ns name; do
  kubectl delete httproute "$name" -n "$ns"
done
```

Three rollback levels:

Level	Action	Latency
L1	Patch weights: nginx=100, envoy=0	< 2 s
L2	Delete HTTPRoute (xDS pushes removal)	< 5 s
L3	Delete Gateway (nginx handles 100%)	< 5 s

Operational commands:

```
make rollback-install # deploy CronJob + RBAC + PrometheusRule
make rollback-test    # manual trigger (DRY_RUN=true to preview)
make canary-status    # weight split across all canary HTTPRoutes
make chaos             # inject 10% 500s -> watch watchdog fire automatically
```

Annotation Translation

Each of 100 Ingresses carries different annotations. Manual per-route translation does not scale.

make translate-annotations runs two steps:

1. **ingress2gateway** (official SIG tool — [sigs.k8s.io/ingress2gateway](https://github.com/kubernetes-sigs/ingress2gateway)) reads live Ingresses and emits HTTPRoute manifests for common annotations
2. **scripts/translate-annotations.sh** covers the remainder — emits BackendTrafficPolicy + SecurityPolicy stubs per Ingress, flags hard cases

nginx annotation	Envoy Gateway CRD	Field
proxy-read-timeout	BackendTrafficPolicy	timeout.http.requestTimeout
proxy-next-upstream-tries	BackendTrafficPolicy	retry.numRetries
limit-rps	BackendTrafficPolicy	rateLimit.local
whitelist-source-range	SecurityPolicy	ipAllowList.cidrRanges
auth-url	SecurityPolicy	extAuth.http
rewrite-target	HTTPRoute filter	URLRewrite.path
server-snippet / modsecurity	EnvoyExtensionPolicy	Wasm — manual work required

Each policy attaches via targetRef scoped to its own HTTPRoute — 100 routes stay independent.

```
make install-ingress2gateway # install official SIG tool
make annotation-audit       # fleet-wide inventory of every annotation
make translate-annotations  # generate HTTPRoutes + policy stubs -> manifests/generated/
```

Failure Injection + Demo

```
make up # bootstrap full cluster (~4 min on MacBook)
make chaos # 10% random 500s -> watchdog auto-rollback fires
make bad-traffic # hammer /fail -> visible error spike
make slow-traffic # hammer /slow?seconds=2 -> p99 breach
make demo-step1 # through demo-step6: full live walkthrough
```